

[Subject code for Theory-2611001305033001]

[Subject code for Practical-2611001305033002]

VEER NARMAD SOUTH GUJARAT UNIVERSITY, SURAT
B.Sc.(Data Science and Analytics) (Semester-5)
SYLLABUS

DS-505: Big Data Handling and Management for Machine Learning Applications

Program Name	Bachelors of Science in Data Science and Analytics								
Semester	5								
NCrF Credit Level	5.5								
Course Type	Major								
Course Subtype	Employability								
Subject Type	Discipline Specific								
Course Code	DS-505								
Course Level	400-499								
Course Title	Big Data Handling and Management for Machine Learning Applications								
Credit	4 Credit								
Effective From	Academic Year : 2026-2027								
Course Outcomes	CO1: Explain the fundamental concepts of Big Data and identify the need for handling large datasets in Machine Learning and AI applications. CO2: Apply Python libraries such as NumPy and Pandas to load, explore, and manage large datasets using essential data processing functions. CO3: Demonstrate the use of PySpark to perform distributed data processing and execute basic operations on large datasets. CO4: Apply data preprocessing and feature engineering techniques to prepare large datasets for Machine Learning models using Scikit-learn. CO5: Analyze and implement basic text data processing and simple LLM-based applications using Python and pretrained APIs.								
Mapping between Cos and PSOs		PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8
	CO1								
	CO2								
	CO3								
	CO4								
	CO5								
Course Content	Unit – 1: Introduction to Big Data and Large Dataset Handling 1. Big Data Fundamentals 1.1 Introduction to Big Data 1.1.1 Definition and characteristics of Big Data (Volume, Velocity, Variety) 1.1.2 Need for Big Data in Machine Learning and AI applications 1.1.3 Examples of large datasets (social media, sensor data, text datasets) 1.2 Working with Large Datasets using Python								

1.2.1 Loading large datasets using Python
1.2.2 Working with CSV, JSON, and Excel datasets
1.2.3 Memory management concepts while handling large datasets
1.3 Essential Python Libraries
1.3.1 Introduction to NumPy and Pandas for large dataset processing
1.3.2 DataFrames and Series concepts
1.3.3 Data exploration techniques
1.4 Practical Commands and Functions
1.4.1 Pandas Functions : read_csv(), read_json(), head(), tail(), info(), describe(), shape, columns
1.4.2 Data Handling functions: drop(), dropna(), fillna(), rename()
Unit – 2: Big Data Processing using PySpark
2.1 Distributed Data Processing
2.2 Introduction to Distributed Processing
2.2.1 Limitations of traditional data processing
2.2.2 Concept of distributed data processing
2.2.3 Introduction to Apache Spark
2.3 Apache Spark Basics
2.3.1 Spark architecture overview
2.3.2 Spark DataFrames
2.3.3 Advantages of Spark for large datasets
2.4 Working with PySpark
2.4.1 Installing PySpark using pip
2.4.2 Creating Spark sessions
2.4.3 Loading large datasets
2.5 Practical Commands and Functions
[Unit-2: Suggested practical : Students should practice: Installation of spark session, ,load dataset and practic Basic operations like: df.show(), df.printSchema(), df.select(), df.filter(), df.groupBy(), df.count()]
Unit – 3: Preparing Big Data for Machine Learning
3. Data Preparation for ML
3.1 Data Cleaning
3.1.1 Handling missing values
3.1.2 Removing duplicate records
3.1.3 Data normalization and scaling
3.2 Feature Engineering
3.2.1 Selecting relevant features
3.2.2 Encoding categorical data
3.2.3 Splitting datasets for training and testing
3.3 Working with Machine Learning Libraries
3.3.1 Introduction to Scikit-learn
3.3.2 Using datasets for ML models
3.4 Practical Commands and Functions
3.4.1 Scikit-learn Functions:train_test_split(),StandardScaler(), LabelEncoder() fit(), predict()
3.4.2 Example Models
3.4.3 LinearRegression
3.4.4 DecisionTreeClassifier
3.4.5 KNeighborsClassifier
Unit – 4: Introduction to Large Language Models and Big Data Applications
4.1 Introduction to LLM
4.1.1Concept of Large Language Models
4.1.2 Examples of LLM applications (chatbots, text summarization)

	4.1.3 Importance of large datasets for LLM training 4.2 Text Data Processing 4.2.1 Working with large text datasets 4.2.2 Tokenization and text cleaning 4.2.3 Basic Natural Language Processing concepts 4.3 Using Pretrained LLM APIs 4.3.1 Introduction to HuggingFace models 4.3.2 Using APIs for text generation 4.3.3 Ethical considerations in AI systems 4.4 Practical Implementation and important Functions 4.4.1 Installation of transformers 4.4.2 Text Processing using Python : split(), lower(), replace(), count() [Tools recommended : Python, NumPy, Pandas, PySpark, Scikit-learn, HuggingFace Transformers, Google Colab / Jupyter Notebook]
Reference Books	1. Big Data: Principles and Best Practices of Scalable Real-Time Data Systems, Nathan Marz and James Warren, ISBN: 9781617290341 2. Hadoop: The Definitive Guide, Tom White, ISBN: 9781491901632 3. Learning Spark: Lightning-Fast Data Analytics, Jules S. Damji, Brooke Wenig, Tathagata Das and Denny Lee, ISBN: 9781492050049 4. Python for Data Analysis, Wes McKinney, ISBN: 9781491957660 5. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Aurélien Géron, ISBN: 9781492032649 6. Data Science from Scratch: First Principles with Python, Joel Grus, ISBN: 9781492041139 7. Machine Learning with PySpark, Tomasz Drabas and Denny Lee, ISBN: 9781784390796 8. Natural Language Processing with Transformers, Lewis Tunstall, Leandro von Werra and Thomas Wolf, ISBN: 9781098103247 9. Mastering Large Datasets with Python, John Wolohan, ISBN: 9781617295285 10. Big Data Analytics with Spark: A Practitioner's Guide to Using Spark for Large Scale Data Analysis, Mohammed Guller, ISBN: 9781484206487
Teaching Methodology	Class Work, Discussion, Self-Study, Seminars and/or Assignment, continuous evaluation and Internal Practicals.
Evaluation Method	Internal Assessment : - Continuous evaluation, Assignment, Attendance - 50% weightage for Practical and Theory exams External Assessment : - Theory exam - Practical exam - 50% weightage for Practical and Theory exams